Fear-Free Night Navigator · The Big Code 2026 Hackathon

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GitHub Repository	https://github.com/ap5967ap/bigcode/
README	https://github.com/ap5967ap/bigcode/blob/main/Aditya%20Priyadarshi_The%20Big%20Code_2026.md

BRIEF SUMMARY

NightSafe Routes is an AI-assisted navigation prototype for Bangalore that treats **psychological safety** as a first-class routing objective for night travel. Rather than optimising purely for ETA, the system assigns each road segment a Safety Score derived from geospatial proxy signals — lighting intensity, commercial activity, street connectivity, transit proximity, and dead-end risk — then exposes four route variants: **Fastest, Balanced, Safest,** and a **personalised Agent Route**. The prototype integrates a FastAPI backend, a React + Leaflet map interface, SHAP-based segment-level explanations, and a Pareto-frontier visualisation of ETA vs. safety trade-offs.

PROBLEM STATEMENT

Standard navigation apps optimise almost exclusively for minimum travel time, frequently routing users through dark, isolated, or low-activity streets to save a few minutes. For solo travellers and women commuting at night, the shortest path can be deeply anxiety-inducing — and may drive costly behavioural changes such as avoiding trips or paying for taxis unnecessarily.

NightSafe Routes addresses the **Fear-Free Night Navigator** challenge: generating routes that are not merely shorter but safer to traverse under nighttime conditions, prioritising peace of mind for solo commuters, women riders, students, and other vulnerable road users.

DESIGN IDEA AND APPROACH

Pipeline Overview

- Load a Bangalore road graph and POI data from OpenStreetMap (OSMnx).
- Engineer edge-level safety proxy features (lighting, activity, connectivity, transit proximity, dead-end penalty).
- Generate synthetic training labels that vary by time-of-day (default / evening / night).
- Train three XGBoost regressors to predict time-aware safety scores per segment.
- Train an MLP archetype classifier to map user context (travel mode, hour, gender, destination type) to one of four routing archetypes.
- Train per-archetype route agents with PPO (Stable-Baselines3); fall back to Q-learning when throughput is insufficient.
- Serve route variants and SHAP-derived segment explanations via a FastAPI + React frontend.

Technologies

- **Backend:** Python · FastAPI · Uvicorn · OSMnx · NetworkX · GeoPandas · Shapely
- **ML / RL:** XGBoost · Scikit-learn MLP · Stable-Baselines3 PPO · Q-learning · SHAP
- **Frontend:** React 18 · Vite · Leaflet · Recharts

Scaling & Rollout

The dominant scaling parameters are the number of road segments, concurrent route requests, and RL agent training time over large subgraphs. The design pre-computes and caches processed graphs and trained models, serving routes from an in-memory graph to keep per-request latency low (mean 414 ms, p95 730 ms). A staged rollout begins with a single Bangalore corridor at night-only hours, collects user comfort feedback, then progressively adds richer real-world supervision (lighting surveys, incident reports) before expanding to further neighbourhoods and cities.

Privacy & Security

The prototype uses lightweight, ephemeral user context (travel mode, hour, destination type, female-rider flag). A production deployment would minimise retained personal data, avoid storing trip histories by default, and enforce strict consent and deletion policies for any sensitive attributes.

PROTOTYPE SNAPSHOT

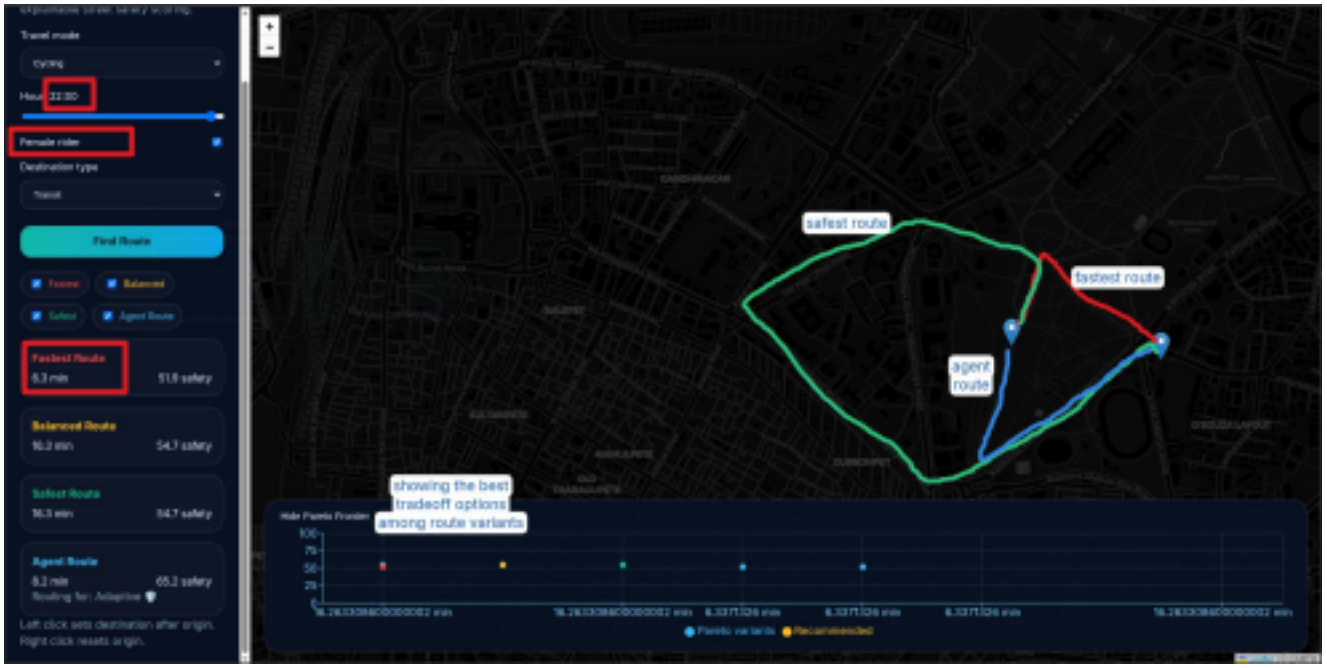

Figure 1 — NightSafe Routes UI showing four route variants on a dark Bangalore basemap, route summary cards (ETA + mean safety score), and the Pareto-frontier scatter plot.

IMPACT

Many people — especially women and solo travellers — change behaviour, spend extra money on cabs, or avoid trips altogether because standard maps ignore night-time safety concerns. NightSafe Routes addresses this gap by internalising psychological safety as a measurable routing objective.

- Safer recommended paths for vulnerable users, grounded in concrete urban signals.
- Higher confidence in night travel through explicit ETA vs. safety trade-off control.
- Improved transparency via SHAP-derived segment explanations (e.g. 'isolated dead-end segment' or 'near transit stops').
- Modular architecture allowing progressive improvement as stronger real-world labels (incident data, lighting surveys) become available.

FEASIBILITY

The implementation is feasible because a working vertical slice is already in place:

- Processed geospatial graph artifacts, trained XGBoost models, RL agents, and SHAP explanations are all versioned

in the repository.

• Backend and frontend components are integrated and tested with automated test suites covering the API, routing logic, pipeline, and RL environment.

• The synthetic label strategy — using high-quality proxy signals when real-world incident data is unavailable — is explicitly encouraged in the problem statement and produces credible scores.

• The modular design allows stronger supervision to be swapped in later without redesigning the system.

USE OF AI

AI is central to every layer of the solution, not merely decorative:

• **XGBoost regression** — infers edge-level safety from seven geospatial proxy features with Optuna-tuned hyperparameters.

• **SHAP explainability** — provides segment-level feature attributions rendered directly in the frontend. • **MLP archetype classifier** — maps user context to one of four routing archetypes (Vulnerable Solo, Comfort Seeker, Efficiency First, Adaptive).

• **PPO reinforcement learning** (with Q-learning fallback) — learns archetype-specific route behaviour from an OpenAI Gym-compatible environment.

DEMONSTRABLE RELIABILITY

Safety Scoring Models

Default (day)	3.08	2.46
Evening model	3.03	2.42
Night model	3.01	2.39

• **Archetype classifier accuracy:** 59.9 %

• **Full-model mean safest-route safety:** 76.57 / 100

• **Route latency — mean / p95 / p99:** 414 ms / 730 ms / 989 ms

Feature-importance rankings consistently highlight **dead-end penalty** and **activity score** as dominant signals. Ablation studies confirm that removing activity score, connectivity score, or time-aware scoring each degrades safest-route mean safety — validating the design assumptions. Notably, the personalised agent route currently underperforms the heuristic safest route in offline evaluation; this honest result surfaces a clear next step — improved reward shaping and extended policy training — rather than masking it.

ALTERNATIVES CONSIDERED

• **Rule-only scoring** — Simpler but too brittle and hard to calibrate across different neighbourhood types. • **End-to-end deep route recommendation** — Less interpretable and unrealistic given the limited labelled trajectory data available.

• **Crime-data-only routing** — Attractive in theory, but crime datasets are often sparse, delayed, and poorly aligned to block-level perception.

• **Purely personalised routing without a shared safety model** — Weaker transparency and harder to justify in high-trust mobility settings.

REFERENCES & APPENDICES

• **OpenStreetMap** — road network and POI data (downloaded via OSMnx).

• **Repository artifacts:** data/evaluation_report.json (offline metrics), data/processed/training_curves.json (RL training snapshots), data/processed/shap_explanations.json (per-segment SHAP values).

• **API:** FastAPI endpoints — POST /route (route variants), GET /segment/{edge_id}/explain (SHAP explanation), GET /health (graph stats).

• **Frontend:** React 18 + Leaflet — dark basemap, four overlaid colour-coded route polylines, segment tooltip on hover, Pareto scatter chart.

• **Tests:** tests/test_api.py, test_classifier.py, test_pipeline.py, test_rl_env.py, test_router.py — covering all major subsystems.

Before final submission, replace the participant placeholders in the metadata table and insert the public GitHub repository link.